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Titolo

Macchine elettriche rotanti

Parte 14: Vibrazioni meccaniche di macchine con altezza d'asse uguale o superiore a 56 mm

Misura, valutazione e limiti della intensità di vibrazione

Title

Rotating electrical machines

Part 14: Mechanical vibration of certain machines with shaft heights 56 mm and higher

Measurement, evaluation and limits of vibration

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COMITATO ELETTROTECNICO ITALIANO

CNR CONSIGLIO NAZIONALE DELLE RICERCHE • **AEI** ASSOCIAZIONE ELETTROTECNICA ED ELETTRONICA ITALIANA

SOMMARIO

La Norma specifica le condizioni di prova e di misura e fissa i limiti del livello di intensità di vibrazione di una macchina elettrica, quando le misure vengano eseguite sulla macchina disaccoppiata, in un'area di prova e in condizioni adeguatamente controllate.
La presente EN viene pubblicata dal CEI nella sola lingua originale inglese, a causa della limitata utilizzazione, particolarmente mirata a settori specialistici.

DESCRITTORI • *DESCRIPTORS*

Macchine elettriche rotanti • *Rotating electrical machine;* Vibrazione • *Vibration;* Prova • *Test;* Condizione di prova • *Test condition;* Condizione di misura • *Measurement condition;* Severità della vibrazione • *Vibration severity;* Valore massimo • *Maximum value;*

COLLEGAMENTI/RELAZIONI TRA DOCUMENTI

Nazionali	(UTE) CEI EN 60034-1:1996-05;
Europei	(IDT) EN 60034-14:1996-12;
Internazionali	(IDT) IEC 60034-14:1996-11;
Legislativi	

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LEGENDA
(UTE) La Norma in oggetto deve essere utilizzata congiuntamente alle Norme indicate dopo il riferimento (UTE)
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Europäische Norm • Norme Européenne • European Standard • Norma Europea

EN 60034-14:1996-12

Sostituisce il documento HD 53.14 S1 (1992)

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Parte 14: Vibrazioni meccaniche di macchine con altezza d’asse uguale o superiore a 56 mm
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Up-to-date lists and bibliographical references concerning such National Standards may be obtained on application to the Central Secretariat or to any CENELEC member.

This European Standard exists in three official versions (English, French, German).

A version in any other language and notified to the CENELEC Central Secretariat has the same status as the official versions.

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C E N E L E C

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CONTENTS

FOREWORD

The text of document 2/940/FDIS, future edition 2 of IEC 34-14, prepared by IEC TC 2, Rotating machinery, was submitted to the IEC-CENELEC parallel vote and was approved by CENELEC as EN 60034-14 on 1996/10/01.

This European Standard supersedes HD 53.14 S1:1992.

The following dates were fixed:

- latest date by which the EN has to be implemented at national level by publication of an identical national standard or by endorsement
  (dop) **1997/08/01**
- latest date by which the national standards conflicting with the EN have to be withdrawn
  (dow) **1997/08/01**

For products which have complied with HD 53.14 S1:1992 before 1997/08/01, as shown by the manufacturer or by a certification body, this previous standard may continue to apply for production until 2002/08/01.

Annexes designated “normative” are part of the body of the standard.

Annexes designated “informative” are given for information only.

In this standard, annex ZA is normative and annex A is informative.

Annex ZA has been added by CENELEC.

ENDORSEMENT NOTICE

The text of the International Standard IEC 34-14:1996 was approved by CENELEC as a European Standard without any modification.

1 SCOPE AND OBJECT

This part of IEC 34 specifies the vibration test procedures and limits for certain electrical machines under specified conditions, when uncoupled from any load or prime mover.

It is applicable to d.c. and three-phase a.c. machines, with shaft heights 56 mm and higher and a rated output up to 50 MW, at nominal speeds from 600 rev/min up to and including 3600 rev/min.

For vertical machines and flange-mounted machines, this standard is only applicable to those machines that are measured in a state of free suspension.

This standard is not applicable to machines mounted in situ, three-phase commutator motors, single-phase machines, three-phase machines operated on single-phase systems, vertical water power generators, permanent magnet generators, or series-wound d.c. machines.

Note *For machines measured in situ refer to ISO 10816-1 (basic document).*

2 NORMATIVE REFERENCES

The following normative documents contain provisions which, through reference in this text, constitute provisions of this part of IEC 34. At the time of publication, the editions indicated were valid. All normative documents are subject to revision, and parties to agreements based on this part of IEC 34 are encouraged to investigate the possibility of applying the most recent editions of the normative documents indicated below(1). Members of IEC and ISO maintain registers of currently valid International Standards.

3 DEFINITIONS

For the purpose of this standard, the definitions in the normative references listed in clause 2 apply.

(1) **Editor's Note:** For the list of Publications, see Annex ZA.

4 MEASUREMENT QUANTITIES

4.1 General

Measurement quantities are the vibration velocity at the machine bearings and the relative shaft vibration displacement within or near to the machine bearings.

4.2 Vibration severity

The criterion for the vibration at the machine bearings shall be the r.m.s. value of the vibration velocity in millimetres per second. The greatest value, determined at the prescribed measurement points, characterizes the vibration severity of the machine (see ISO 10816-1).

Induction motors (especially of two-pole type) frequently show vibration velocities beating at twice slip frequency. In these cases the vibration severity shall be determined from the relationship:

$$v_{\text{r.m.s.}} = \sqrt{\frac{(v_{\max}^2 + v_{\min}^2)}{2}}$$

where

$v_{\max}$ is the maximum r.m.s. velocity value;

$v_{\min}$ is the minimum value.

4.3 Relative shaft vibration

The criterion adopted for the relative shaft vibration shall be the vibratory displacement $S_{p\text{-}p}$ in the direction of measurement (see ISO/DIS 7919-1).

5 MEASUREMENT EQUIPMENT

Measurement equipment for vibration severity measurements shall comply with the requirements in ISO 2954.

Measurement equipment for relative shaft vibration measurements shall comply with the requirements in ISO/DIS 10817-1.

6 MACHINE MOUNTING

The vibration of an electrical machine is closely linked with the mounting of the machine. To permit evaluation as far as balance and vibration of rotating electrical machines are concerned, it is necessary to measure the vibration on the machine alone, under properly determined test conditions, to enable reproducible tests to be carried out and to provide comparable measurements.

6.1 Free suspension

This condition is achieved by suspending the machine on a spring or by mounting on an elastic support (springs, rubber, etc.).

The natural oscillation frequency of the suspension system and machine, in the six possible degrees of freedom, shall be less than 1/3 of the frequency corresponding to the speed of the machine under test, as defined in 7.3. The necessary elasticity of the suspension system, as a function of nominal speed, can be determined from figure 1.

6.2 Rigid mounting

The machine shall be clamped firmly to a solid floor.

The maximum vibration velocity measured in the horizontal and vertical directions at the machine feet (or at the base frame near to the bearing pedestals or stator feet) shall not exceed 25% of the maximum velocity which is measured at the adjacent bearing in either the horizontal or vertical direction.

Notes: 1 *This requirement ensures that the horizontal and vertical natural frequencies of the complete test arrangement do not coincide within:*
a ± 10% of the rotational frequency of the machine;
b ± 5% of two times rotational frequency;
c ± 5% of one and two times electrical line frequency.

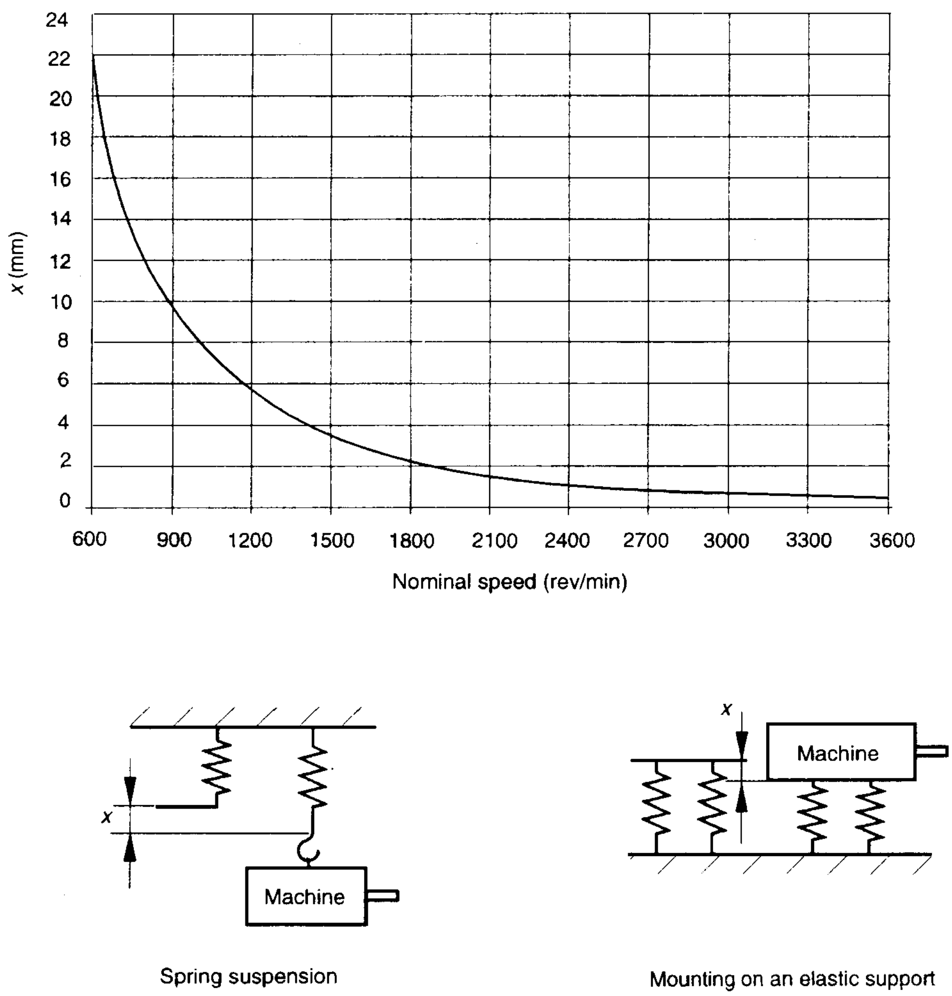
2 *The ratio of 25% of foot to bearing vibration velocities is valid for once per revolution vibration and the vibration at twice line frequency (if the latter is being evaluated).*

6.3 Active environment determination

The support systems described in 6.1 and 6.2 are considered passive, admitting insignificant external disturbances to the machine. If the vibration with the machine stationary exceeds 25% of the value when the machine is running, then an active environment is said to exist and this standard does not apply (see ISO 10816-1).

Fig. 1 **Minimum elastic displacement as s function of nominal speed**

Note *To reduce the influence of the mass and the moments of inertia of the suspension system on the vibration level the effective mass of the elastic support should not be greater than 1/10 of that of the machine.*



7 CONDITIONS OF MEASUREMENT

7.1 Key

For the balancing and measurement of vibration on machines provided with a shaft extension keyway, the keyway shall be filled with a half key.

Note *A full length rectangular key of half height or a half length key of full height (which should be centred axially in the keyway) is acceptable (see 3.3 of ISO 8821).*

7.2 Measurement points

7.2.1 Measurement points for vibration velocity

The location of the measurement points and directions to which the levels of vibration severity apply are shown in figure 2 for machines with end-shield bearings and in figure 4 for machines with pedestal bearings. Figure 3 applies to those machines where measurement positions according to figure 2 are not possible without disassembly of parts.

7.2.2 Measurement points for relative shaft displacement

Non-contacting transducers shall be installed inside the bearing, measuring directly the relative shaft journal displacement, or (when inside mounting is not possible) adjacent to the bearing shell. The preferred radial positions are as indicated in figure 5.

7.3 Test conditions

Machines shall be tested at no-load with all relevant quantities at their rated value.

AC machines shall be run with a virtually sinusoidal wave form according to 6.2 of IEC 34-1.

Unless otherwise agreed, the test shall be performed at each rated speed or over the whole range of rated speeds. For each tested speed the values shall not exceed the corresponding limit in table 1.

For machines with bidirectional rotation, the vibration limits apply for both directions.

7.4 Vibration transducer

The mounting of the transducer used for vibration velocity measurement and the machine surface shall be as specified by the manufacturer of the transducer and shall not disturb the vibratory condition of the machine under test.

For this it is necessary that the total coupled mass of the transducer assembly is less than 1/50 of the mass of the machine.

8.1 Limits of vibration severity

The limit of vibration severity for d.c. and three-phase a.c. machines with shaft heights 56 mm and higher and for the two mounting conditions as specified in clause 6 are specified in table 1. Values are given for three vibration grades: N (normal), R (reduced) and S (special).

Tab. 1 **Limits of vibration severity in mm/s r.m.s. for the shaft height *H* (mm)**

		Machine measured in a state of free suspension				Rigid mounted
Vibration grade	Rated speed rev/min	56< $H \leq$ 132	132< $H \leq$ 225	225< $H \leq$ 400	H > 400	H > 400
N	600 to 3 600	1,8	2,8	3,5	3,5	2,8
R	600 to 1800	0,71	1,12	1,8	2,8	1,8
	>1800 to 3600	1,12	1,8	2,8	2,8	1,8
S	600 to 1800	0,45	0,71	1,12		
	>1800 to 3600	0,71	1,12	1,8		

Notes:

1. If no grade is specified, machines complying with this standard shall be grade N.
2. Grade R machines are frequently ordered for machine-tool drives. Grade S machines are in use as drives for special machines with extreme vibration requirements. Grade S is applicable only to machines with shaft heights ≤400 mm.
3. For machines requiring values lower than those given in table 1, it is recommended to select values from the series 0,45; 0,71: 1,12; 1,8 and 2,8 mm/s.
4. For machines with H > 400 mm, both methods of mounting are applicable. Unless otherwise agreed, the choice of the method will be made by the manufacturer.
5. Manufacturer and purchaser should take into account that the instrumentation can have a measurement tolerance of ±10%.
6. A machine, which is well balanced in itself and of a grade conforming with the table, may exhibit large vibrations, when installed *in situ*, arising from various causes, such as unsuitable foundations, reaction of the driven machine, etc. Vibration may also be caused by driving elements with a natural oscillation frequency very close to the excitation due to the small residual unbalance of the rotating masses of the machine. In such cases, checks should be carried out not only on the machine, but also on each element of the installation.
7. The shaft height of a machine without feet, or a machine with raised feet, or any vertical machine is to be taken as the shaft height of a machine in the same basic frame, but of the horizontal shaft, loot-mounting type.8.2 Limits of vibration velocity with twice line frequency for a.c. machines

8.2 Limits of vibration velocity with twice line frequency for a.c. machine

Two-pole machines may have significant vibration at twice the frequency of the power system. The correct evaluation of these vibration components requires a rigid mounting of the machine which complies with the requirements given in 6.2.

For two-pole machines with shaft heights H > 225 mm, the above evaluation shall be part of the basic machine type test to establish the existence of significant twice line frequency vibration.

Machines having H > 225 mm and twice line frequency vibration, shall be tested with rigid mounting and shall not exceed the values for H > 400 with rigid mounting in table 1.

8.3 Axial vibration

The evaluation of axial bearing vibration depends on the bearing function and bearing construction.

In the case of thrust bearings, axial vibration correlates with thrust pulsations which could cause damage to the metal liner of sleeve bearings or to parts of antifriction bearings. Axial vibration of these bearings shall be evaluated in the same manner as transverse vibration and table 1 limits apply.

Where bearings have no axial limiting construction, a less exacting requirement may be acceptable, subject to prior agreement between manufacturer and purchaser.

9 LIMITS OF RELATIVE SHAFT VIBRATION

Relative shaft vibration measurements are recommended only for two- and four-pole machines with sleeve bearings and rated power >1000 kW and shall be the subject of prior agreement between manufacturer and purchaser with respect to the necessary provisions regarding the installation of shaft measurement transducers.

When machines with sleeve bearings have special provisions for the installation of shaft vibration measurement transducers, the limits for the relative shaft vibration displacement are specified in table 2. These limits are in addition to those required in clause 8.

Tab. 2 **Limits for the maximum shaft vibration (S_{p-p}) and the maximum run-out**

Vibration grade	Number of poles	Maximum relative shaft displacement µm	Maximum run-out µm
N	2	70	18
	4	90	23
R	2	50	12,5
	4	70	18

Notes:
1. Machines with vibration grade R are frequently specified for high-speed drives in critical installations.
2. All limits apply to both 50 Hz and 60 Hz machines.
3. The maximum relative shaft displacement limits include the run-out. For the definition of the run-out (see ISO/DIS 7919-1).

Fig. 2 **Preferred points of measurement applicable to one or both ends of the machine**

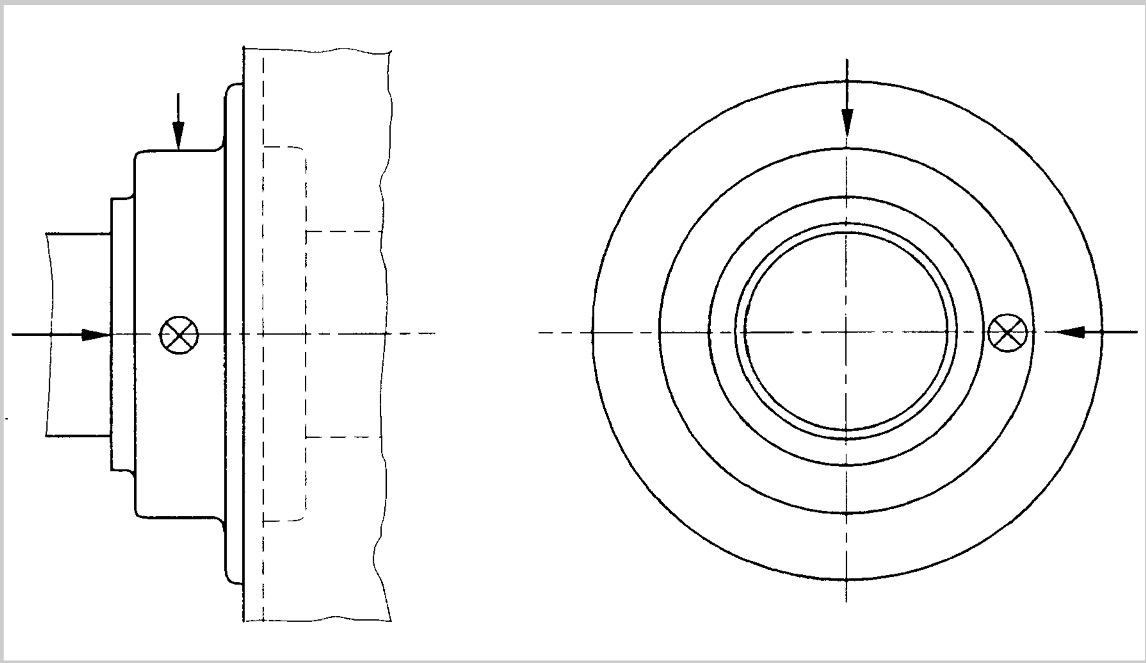


Fig. 3 **Measurement points for those ends of machines where measurements according to figure 2 are not possible without disassembly of parts**

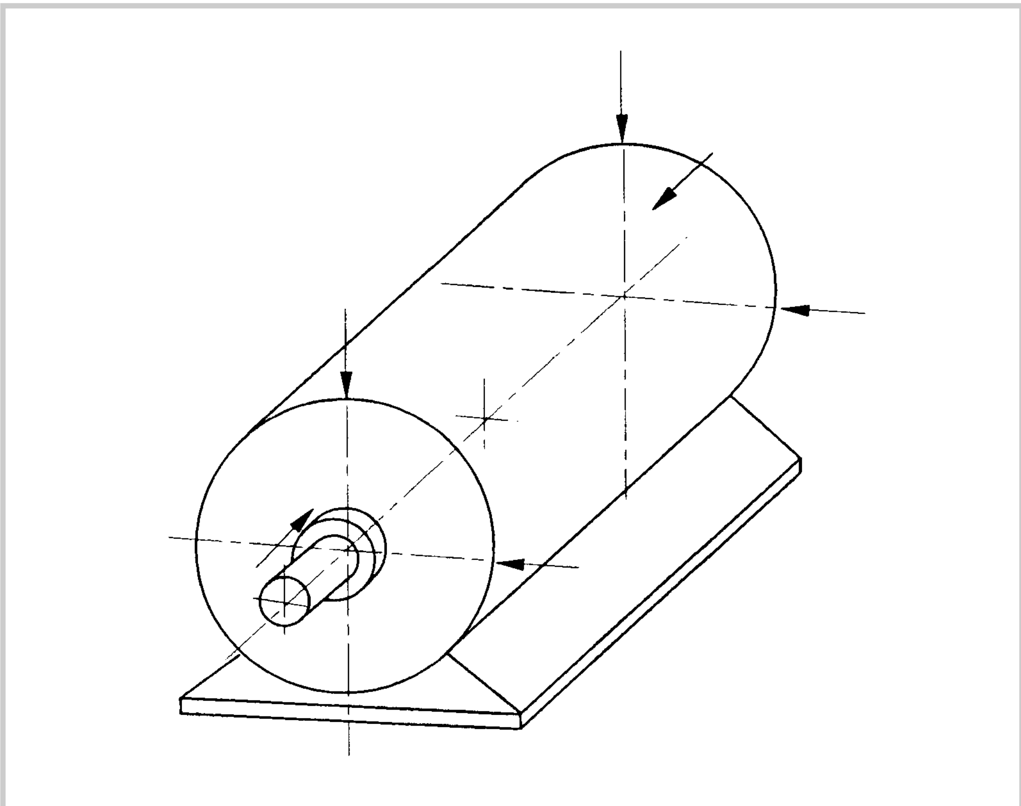


Fig. 4 **Measurement points for pedestal bearing**

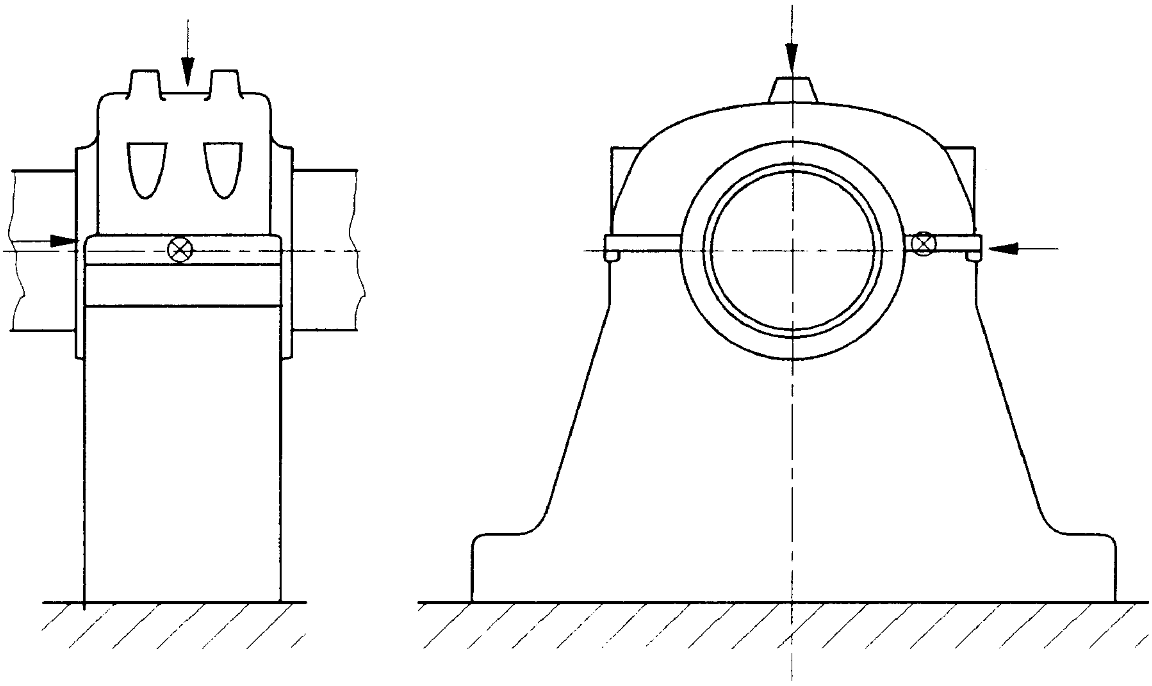
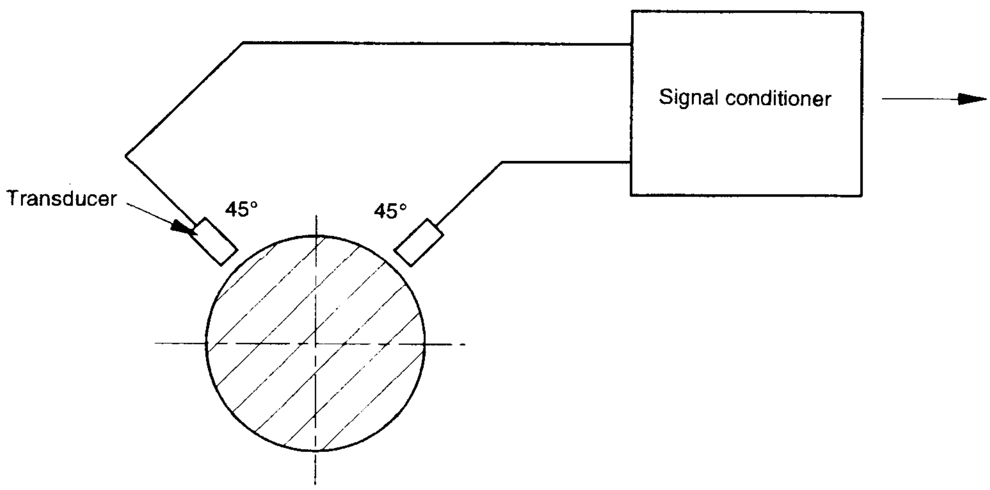


Fig. 5 **Preferred circumferential position of transducers for the measurement of relative shaft displacement**



ANNEX A informative

BIBLIOGRAPHY

ISO/DIS 7919-3, *Mechanical vibration of non-reciprocating machines - Measurements on rotating shafts and evaluation - Part 3: Guidelines for coupled industrial machines*

API Standard 541: 1987, *Form-wound squirrel-cage induction motors 250 horsepower and larger*

API Standard 546: 1990, *Form-wound brushless synchronous motors 500 horsepower and larger*

ANNEX ZA normative

Normative references to international publications with their corresponding European publications

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

Note *When an international publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.*

Publication	Year	Title	EN/HD	Year
IEC 34-1	1996(1)	Rotating electrical machines Part 1: Rating and performance	—	—
ISO 2954	1975	Mechanical vibration of rotating and reciprocating machinery - Requirements for instruments for measuring vibration severity	—	—
ISO/DIS 7919-1		Mechanical vibration of non-reciprocating machines - Measurements on rotating shafts and evaluation Part 1: General guidelines	—	—
ISO 8821	1989	Mechanical vibration - Balancing Shaft and fitment key convention	—	—
ISO 10816-1	1995	Mechanical vibration - Evaluation of -machine vibration by measurement on non-rotating parts Part 1: General guidelines	—	—
ISO/DIS 10817-1		Rotating shaft vibration measuring system Part 1: Relative and absolute signal sensing of radial vibration from rotating shafts	—	—

(1) IEC 34-1:1994, mod., is harmonized as EN 60034-1:1995.

Fine Documento

La presente Norma è stata compilata dal Comitato Elettrotecnico Italiano
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Responsabile: Ing. E. Camagni

2 – Macchine rotanti

CEI EN 60034-1 (CEI 2-3)
Macchine elettriche rotanti Parte 1: Caratteristiche nominali e di funzionamento

CEI EN 60034-4 (CEI 2-5)
Macchine elettriche rotanti Parte 4: Metodi per determinare dalle prove le grandezze delle macchine sincrone

CEI 2-6
Macchine elettriche rotanti - Metodi di determinazione, mediante prove, delle perdite e del rendimento

CEI EN 60034-6 (CEI 2-7)
Macchine elettriche rotanti Parte 6: Metodi di raffreddamento (Codice IC)

CEI 2-8
Macchine elettriche rotanti Parte 8: Marcatura dei terminali e senso di rotazione delle macchine rotanti

CEI 2-9
Macchine con tensione nominale compresa fra 5 kV e 24 kV Prove di isolamento delle barre e delle matasse

CEI 2-10
Metodi di prova per la misura delle caratteristiche fisiche dei materiali delle spazzole per macchine elettriche

CEI 2-13
Macchine elettriche rotanti - Misura delle perdite con il metodo calorimetrico

CEI EN 60034-7 (CEI 2-14)
Macchine elettriche rotanti Parte 7: Classificazione delle forme costruttive e dei tipi di installazione

CEI EN 60034-12 (CEI 2-15)
Macchine elettriche rotanti Parte 12: Caratteristiche di avviamento dei motori asincroni trifase a gabbia, ad una sola velocità, a 50 Hz e per tensioni di alimentazione inferiori o uguali a 690 V

CEI 2-17
Macchine elettriche rotanti Parte 15: Livelli di tensione di tenuta ad impulso delle macchine rotanti a corrente alternata con bobine statoriche preformate.

CEI 2-18
Guida per l'impiego e l'esercizio di macchine sincrone a rotore liscio raffreddate in idrogeno

CEI 2-19
Dimensioni delle spazzole e dei portaspazzole per macchine elettriche

CEI EN 60276 (CEI 2-20)
Definizioni e nomenclatura per spazzole di carbone, portaspazzole, commutatori e collettori ad anello

CEI EN 60034-3 (CEI 2-22)
Macchine elettriche rotanti Parte 3: Prescrizioni specifiche per macchine sincrone a rotore liscio (turboalternatori)

CEI 2-23
Macchine elettriche rotanti Parte 14: Vibrazioni meccaniche di macchine con altezza d'asse uguale o superiore a 56 mm. Misura, valutazione e limiti della intensità di vibrazione

CEI EN 60034-9 (CEI 2-24)
Macchine elettriche rotanti Parte 9: Limiti di rumore

CEI EN 60034-18-1 (CEI 2-25)
Macchine elettriche rotanti Parte 18: Valutazione funzionale dei sistemi di isolamento Sezione 1: Principi direttivi generali

CEI EN 60034-22 (CEI 2-28)
Macchine elettriche rotanti Parte 22: Generatori a corrente alternata per gruppi elettrogeni azionati da motori a combustione interna a pistoni

Lire **62.000**

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